
Christopher Barker

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SELECTED PROJECTS & EVIDENCE

- [Protein AI Pipeline](#) — research poster and public workstream.
- [AI Caption Generator](#) — Streamlit app and GitHub repo.
- [SiDiYa Branding System](#) — identity system and master board.
- [Grocery Retail Consumer Analytics](#) — data visualization evidence.
- [HTML Resume Portfolio](#) — portfolio engineering and responsive design.

Frontend Web Developer | React | UX/UI | Data Visualization

EDUCATION

Pacific Lutheran University

February 2024 - May 2026 | Tacoma, WA

Bachelor of Arts - Individualized major Web Development and Interactive Design

Tacoma Community College

June 2017 - June 2022 | Tacoma, WA

Associate of Arts - Direct Transfer Agreement

TECHNICAL ACTIVITIES

UX/UI | TestUnlimited.com | Intern

January 2024 - March 2024

- Converted sketches into client-ready UI blueprints and high-fidelity deliverables.
- Redesigned product page layouts and visual assets to improve usability and clarity.
- Worked with engineering to deploy front-end updates, ensuring implementation fidelity.

Derailed | React, CSS, JavaScript, JSON

June 2025 - September 2025

- Developed a real-time multiplayer browser game using React and JavaScript, implementing optimized state management for player synchronization.
- Designed responsive game interfaces and controls ensuring smooth gameplay across multiple browsers and devices.
- Implemented optimized state management for seamless player movement and collision detection.

West Coast Tourism Data Analysis | Python, Data Visualization

March 2026

- Engineered a data analysis pipeline to evaluate the "Weekender Effect", comparing coastal versus suburban retail datasets.
- Processed large-scale datasets to determine the statistical impact of regional tourism on premium spending and basket diversity.

AI Protein Sequence Visualization | Python, ESM-3, BLIP-2 June 2025 - Current | Tacoma, WA

- Built a Python-based AI pipeline integrating ESM-3 and BLIP-2 models to convert protein sequences into 2D visual embeddings.

- Generated accurate protein image captions, significantly improving function prediction by mapping sequence embeddings to UniProt databases.

Discord Bot | Python, Git

June 2024 - September 2024

- Developed a functional Python reminder bot utilizing slash commands, implementing countdown timers, and managing version control via GitHub branches.

WORK AND RESEARCH EXPERIENCE

Research | Pacific Lutheran University | Undergraduate Assistant

June 2025 - Current | Tacoma, WA

- Developed a multimodal AI pipeline combining protein embeddings, image captioning, and similarity analysis for protein function inference.
- Contributed to research accepted for presentation at ICIBM 2026 (International Conference on Intelligent Biology and Medicine).

PC Tech | Pacific Lutheran University | Work study

August 2024 - June 2025 | Tacoma, WA

- Troubleshoot diverse hardware/software issues and mentored new student workers to promote cross-functional teamwork.

ACHIEVEMENTS & AWARDS

ICIBM 2026 — Accepted research paper in applied AI and computational biology

2026

Dean's List | University of Washington, Tacoma

June 2024 - Current

First Generation Fellow | University of Washington, Tacoma

June 2023

Outstanding Student Award | Tacoma Community College

2022 - 2023

TECHNICAL SKILLS

Languages: JavaScript, HTML5, CSS3, Python, Java, JSON, R

Frameworks & Libraries: React, Material UI, Node.js, JUnit

Design & Developer Tools: Figma, UI/UX Wireframing, Git, GitHub, VS Code, Docker, IntelliJ